

THE UNIVERSITY OF MELBOURNE

ELEN90098

Reinforcement Learning for Engineering

Workshop 2

Q-Learning

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Task 1

Question 1

The **CliffWalking** environment is a 4×12 grid world. Using (row, column) coordinates, the set of non-terminal states is

$$\mathcal{S} = (\{0, 1, 2\} \times \{0, \dots, 11\}) \cup \{(3, 0)\}.$$

Hence, $|\mathcal{S}| = 37$. The cliff cells $(3, 1), \dots, (3, 10)$ cannot be occupied, while $(3, 11)$ is the terminal goal state. Equivalently, under the integer encoding $s = 12 \text{ row} + \text{column}$, the non-terminal states are $\mathcal{S} = \{0, \dots, 36\}$, with $s = 36$ as the initial state. The state space is therefore finite and discrete.

The action space is

$$\mathcal{A} = \{0, 1, 2, 3\},$$

where the actions correspond to up, right, down and left, respectively. Thus, $|\mathcal{A}| = 4$, so the action space is finite and discrete.

The reward set is

$$\mathcal{R} = \{-1, -100\}.$$

A reward of -1 is received for an ordinary time step, while entering a cliff cell gives a reward of -100 and returns the agent to the initial state. The reward space is therefore also finite and discrete.

Question 2

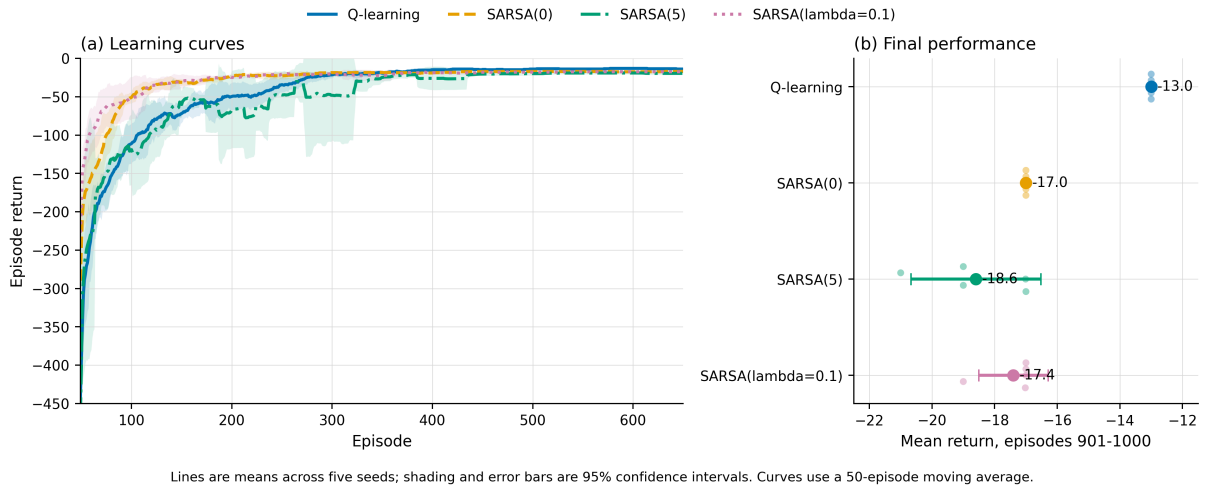


Figure 1: Learning performance of Q-learning, SARSA(0), SARSA(5) and SARSA($\lambda = 0.1$) on the CliffWalking-v1 environment. Panel (a) shows the 50-episode moving-average return over training, averaged across five random seeds, with shaded 95% confidence intervals. Panel (b) shows the mean return over episodes 901–1000 for each seed, together with the cross-seed mean and 95% confidence interval.